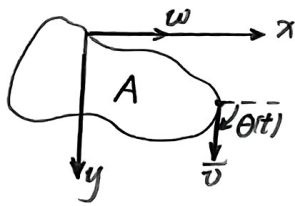


# 最优控制理论

## 第一章 最优控制概论

例1. 渔轮相对于海流的速度为  $\bar{v}$ , 大小不变, 海流速度  $w$  方向大小一定, 求  $\theta(t)$  的最优变化律, 使得在一定  $t_f$  内使得围得面积最大.



解 渔轮运动方程为

$$\begin{cases} \dot{x}(t) = v \cos \theta(t) + w \\ \dot{y}(t) = v \sin \theta(t) \end{cases}$$

初始条件与末端条件

$$\begin{cases} x(0) = x(t_f) = x_0 \\ y(0) = y(t_f) = y_0 \end{cases}$$

最优控制律, 性能指标选为面积:

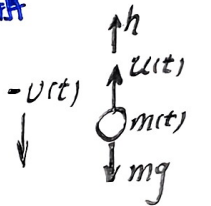
$$J = \oint y(t) dx(t) = \int_0^{t_f} y(t) \dot{x}(t) dt = \int_0^{t_f} y(t) (v \cos \theta(t) + w) dt$$

最大.

例2. 登月舱质量为  $m$ , 高度为  $h$ , 垂直速度为  $v$ , 发动机推力为  $u$ , 初始高度为  $h_0$ , 速度为  $v_0$ , 空载质量为  $M$ , 燃料质量为  $F$ .

求推力控制律使燃料消耗最小.

解



运动方程为

$$\begin{cases} \dot{h}(t) = v(t) \\ \dot{v}(t) = \frac{u(t)}{m(t)} - g \\ \dot{m}(t) = -k u(t) \end{cases}$$

初态与末态条件

$$\begin{cases} h(0) = h_0 \\ v(0) = v_0 \\ m(0) = M + F \end{cases} \quad \begin{cases} h(t_f) = 0 \\ v(t_f) = 0 \end{cases}$$

控制约束  $0 \leq u(t) \leq U_{max}$

$J = m(t_f)$  越大越好

## 最优控制问题

### 最优控制问题的基本组成

① 系统数学模型  $\dot{x} = f(x(t), u(t), t), t \in [t_0, t_f]$

$x(t)$  为状态向量,  $u(t)$  为控制向量.

② 边界条件与目标集

iii 初始时刻  $t_0$  及其初始状态  $x(t_0)$ :  $\rightarrow$  已知.

iii 末端时刻  $t_f$  及其末端状态  $x(t_f)$ :  $\rightarrow$  未知.

$\downarrow$

$\varphi[x(t_f), t_f] = 0 \rightarrow$   $\begin{cases} x_f(t) \text{ 固定, } \varphi(\cdot) \text{ 仅有一点;} \\ x_f(t) \text{ 有约束, } \varphi(\cdot) \text{ 为 } r \text{ 维超曲面;} \\ x_f(t) \text{ 无约束, } \varphi(\cdot) \text{ 为 } r \text{ 维空间.} \end{cases}$

③ 容许控制  $\left\{ \begin{array}{l} \text{范围受限的控制} \rightarrow \text{闭集} \\ \text{范围无限的控制} \rightarrow \text{开集} \end{array} \right. \Rightarrow \text{控制域} \Omega.$

④ 性能指标 — 代价函数  $J.$

· 最优控制的一般提法:

设系统状态方程及其初始条件为:

$$\begin{cases} \dot{x}(t) = f(x(t), u(t), t) \\ x(t_0) = x_0 \end{cases}$$

要求确定最优控制  $u^*(t)$ , 使系统从已知初态  $x(t_0)$  转移到末态  $x(t_f)$ .

使性能指标:

$$J = \varphi(x(t_f), t_f) + \int_{t_0}^{t_f} L(x(t), u(t), t) dt$$

有极值, 同时满足:

ii) 控制约束:  $g(x(t), u(t), t) \geq 0;$

iii) 目标约束:  $\varphi(x(t_f), t_f) = 0.$

性能指标类型

① 积分型指标  $J = \int_{t_0}^{t_f} L(x(t), u(t), t) dt$

→ 拉格朗日问题.

ii) 最短时间控制  $J = \int_{t_0}^{t_f} dt = t_f - t_0 \quad (L = 1)$

iii) 最小能耗控制  $J = \int_{t_0}^{t_f} \sum |u_j(t)| dt \quad (L = \sum |u_j(t)|)$

iiii) 最小能量控制  $J = \int_{t_0}^{t_f} u^T(t) u(t) dt \quad (L = u^T(t) u(t))$

② 末值型指标  $J = \varphi(x(t_f), t_f)$

→ 迈耶尔问题.

③ 复合型指标

→ 波尔扎问题.

ii) 有限时间线性状态调节器:

iii) 有限时间线性跟踪系统.

## 第二章 古典变分法

### 1. 泛函与变分法

#### 线性赋范空间

**距离空间**  $X$  为距离空间:  $\forall x, y \in X$ , 有实函数  $d(x, y)$ :

i)  $d(x, y) \geq 0$ , 且  $x = y \iff d(x, y) = 0$ ;

ii)  $d(x, y) = d(y, x)$ ;

iii)  $d(x, y) \leq d(x, z) + d(y, z)$

$d(x, y)$  为  $X$  上的距离.

**完备距离空间** ① 柯西序列:  $x_n \in X$ , 对  $\forall \varepsilon > 0$ ,  $\exists$  足够大  $N$ , 使  $m, n > N$  时, 有

$$d(x_m, x_n) < \varepsilon$$

则  $x_n$  为  $X$  的柯西序列;

② 若  $X$  中每个柯西序列均收敛于  $X$  中一点, 则  $X$  为完备距离空间.

**开集与闭集** ① 开球:  $S(x_0; r) = \{x \in X \mid d(x, x_0) < r, r > 0\}$

称为以  $x_0$  为中心, 以  $r$  为半径的开球.

② 内点: 对子集  $M \subset X$ , 存在以  $x$  为中心的开球  $S(x; \varepsilon) \subset M$ , 则  $x$  为  $M$  的内点.

③ 开集:  $M$  内的所有点均为内点.

④ 附着点:  $x$  为中心的开球与  $M$  交集非空, 则  $x$  为  $M$  的附着点;

⑤ 闭集:  $M$  的附着点集合为  $M$  的闭包  $\bar{M}$ ,  $M = \bar{M}$  称  $M$  为闭集.

#### 线性赋范空间

① 线性赋范空间: 对线性空间  $R^n$  中每一个元素  $x \in R^n$  可定义范数  $\|x\|$ :

i)  $\|x\| \geq 0$ , 且  $x = 0 \iff \|x\| = 0$ ;

ii)  $\|x+y\| \leq \|x\| + \|y\|$

iii)  $\|\alpha x\| = |\alpha| \|x\|$

② 希尔伯特空间:  $n$  维线性空间  $R^n$  中:

$$\langle x, y \rangle = \sum_{i=1}^n x_i y_i$$

$\Downarrow$

$$\|x\| = \langle x, x \rangle^{\frac{1}{2}}$$

$\Downarrow$

$$d(x, y) = \|x - y\|$$

则  $R^n$  为希尔伯特空间.

③ 控制向量函数空间:  $u(t) = \begin{pmatrix} u_1(t) \\ u_2(t) \\ \vdots \\ u_m(t) \end{pmatrix}$

$$\langle u(t), v(t) \rangle = \int_{t_0}^{t_f} u^T(t) v(t) dt$$

$\Downarrow$

$$\|u(t)\| = \langle u(t), u(t) \rangle^{\frac{1}{2}}$$

## 泛函

- 对某类函数中的所有函数  $x(t)$ ,  $J$  有唯一确定值与之对应, 则  $J$  为依赖于  $x(t)$  的泛函, 记为  $J = J(x(t))$ .

积分是一种泛函  $J(x) = \int_0^1 x(t) dt$

$$x(t) = t \text{ 时, } J(x) = \frac{1}{2}$$

$$x(t) = \cos t \text{ 时, } J(x) = \sin 1$$

**泛函算子**  $R^n$  为  $n$  维线性赋范空间,  $R$  为实数集, 若有

$$y = J(x), \forall x \in R^n, y \in R.$$

则  $J(x)$  称为  $R^n$  到  $R$  的泛函算子.

若: (i)  $J(x_1 + x_2) = J(x_1) + J(x_2)$

(iii)  $J(\alpha x) = \alpha J(x)$

则  $J(x)$  为线性泛函算子.

**定义域与值域** 在  $R^n$  上使  $J$  作用有意义的全体  $x(t)$  称为  $J$  的定义域, 记为  $D(J)$ .

$D(J)$  在  $J$  作用下集合为  $Z(J)$ .

**泛函的连续性**  $J(x)$  为线性赋范空间  $R^n$  的子集  $D(J)$  到  $R$  中的泛函算子, 若对每一个收敛到  $x_0 \in D(J)$  的序列  $x_n \in D(J)$ , 有

$$\lim_{n \rightarrow \infty} J(x_n) = J(x_0)$$

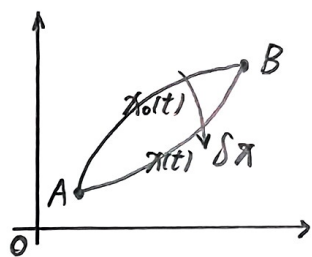
则  $J(x)$  在  $x_0$  连续.

线性泛函必然连续.

## 变分

**函数的变分**  $J(x)$  为连续泛函,  $x(t)$  为函数, 则  $x(t)$  的变分为:

$$\delta x = x(t) - x_0(t)$$



$\delta x$  可引起  $J(x)$  变化:  $J(x + \epsilon \delta x)$ .

$\epsilon = 0$  时, 为原泛函  $J(x)$

$\epsilon = 1$  时, 为增加后泛函  $J(x + \delta x)$

**泛函的变分**  $J(x)$  为线性赋范空间  $R^n$  上的连续泛函, 若其增量表示为.

$$\Delta J(x) = J(x + \delta x) - J(x)$$

$$= L(x, \delta x) + r(x, \delta x)$$

$L(x, \delta x)$  为关于  $\delta x$  的线性连续泛函,  $r(x, \delta x)$  为关于  $\delta x$  的高阶无穷小. 则  $\delta J = L(x, \delta x)$  为  $J(x)$  的变分.



定理(连续泛函的变分)  $J(x)$  为线性赋范空间  $R^n$  上的连续泛函, 若在  $x = x_0$  处  $J(x)$  可微, 则  $J(x)$  在  $x_0$  处的变分为

$$\delta J(x_0, \delta x) = \left. \frac{\partial}{\partial \varepsilon} J(x_0, \varepsilon \delta x) \right|_{\varepsilon=0}$$

○ 若  $J = \int_{t_0}^{t_f} L(x, \dot{x}, t) dt$ , 则  $\delta J = \int_{t_0}^{t_f} \left( \frac{\partial L}{\partial x} \delta x + \frac{\partial L}{\partial \dot{x}} \delta \dot{x} \right) dt$

定理(连续泛函的二次变分)  $J(x)$  为线性赋范空间  $R^n$  上的连续泛函, 若在  $x = x_0$  处二次可微, 则  $J(x)$  在  $x_0$  处的二次变分为

$$\delta^2 J(x_0, \delta x) = \left. \frac{\partial^2}{\partial \varepsilon^2} J(x_0, \varepsilon \delta x) \right|_{\varepsilon=0}$$

## 泛函的极值

• 泛函的极值:  $J(x)$  为线性赋范空间  $R^n$  某个子集  $D$  上的线性连续泛函.  
 $x_0 \in D$ , 若存在  $\delta > 0$ , 使

$$U(x_0, \delta) = \{x \mid \|x - x_0\| < \delta, x \in R^n\}$$

在  $x \in U(x_0, \delta)$  时, 有

$$\begin{cases} \Delta J(x) = J(x) - J(x_0) \geq 0 & (J \text{ 在 } x = x_0 \text{ 时有极小值}) \\ \Delta J(x) = J(x) - J(x_0) \leq 0 & (J \text{ 在 } x = x_0 \text{ 时有极大值}) \end{cases}$$

• 泛函极值的必要条件:  $\delta J(x_0, \delta x) = 0 \iff J(x)$  在  $x = x_0$  有极值.

• 泛函极值的充要条件:  $\begin{cases} \delta J(x_0, \delta x) = 0, \delta^2 J(x_0, \delta x) > 0 \iff J(x) \text{ 在 } x_0 \text{ 有极小值} \\ \delta J(x_0, \delta x) = 0, \delta^2 J(x_0, \delta x) < 0 \iff J(x) \text{ 在 } x_0 \text{ 有极大值} \end{cases}$

• 变分引理: 若  $\zeta(t)$  为  $[t_0, t_f]$  上的向量函数, 对  $n$  维任意连续向量函数.  
 $\eta(t_0) = \eta(t_f) = 0, \int_{t_0}^{t_f} \zeta^T(t) \eta(t) dt = 0$

则  $\zeta(t) \equiv 0$ .

## 2. 变分法

两端固定问题  $x(t_0) = x_0, x(t_f) = x_f$

无约束泛函极值问题  $\min_x J(x) = \int_{t_0}^{t_f} L(x, \dot{x}, t) dt$

取极值的必要条件:  $\frac{\partial L}{\partial x} - \frac{d}{dt} \frac{\partial L}{\partial \dot{x}} = 0$  (Euler方程)

$$L(x^* + \delta x, \dot{x}^* + \delta \dot{x}, t) = L(x^*, \dot{x}^*, t) + \left. \frac{\partial L}{\partial x} \right|_{x^*} \delta x + \left. \frac{\partial L}{\partial \dot{x}} \right|_{\dot{x}^*} \delta \dot{x} + \dots$$

$$\Rightarrow \delta J = \int_{t_0}^{t_f} \left( \left. \frac{\partial L}{\partial x} \right|_{x^*} \delta x + \left. \frac{\partial L}{\partial \dot{x}} \right|_{\dot{x}^*} \delta \dot{x} \right) dt$$

对第二项用分部积分法:  $\delta \dot{x} = \dot{x}(t) - \dot{x}^*(t), \frac{d}{dt} \delta x = \frac{d}{dt} (x(t) - x^*(t)) = \dot{x}(t) - \dot{x}^*(t) = \delta \dot{x}$

$$\int_{t_0}^{t_f} \left. \frac{\partial L}{\partial \dot{x}} \right|_{\dot{x}^*} \delta \dot{x} dt = \left. \left( \left. \frac{\partial L}{\partial \dot{x}} \right|_{\dot{x}^*} \delta x \right) \right|_{t_0}^{t_f} - \int_{t_0}^{t_f} \left( \frac{d}{dt} \left. \frac{\partial L}{\partial \dot{x}} \right|_{\dot{x}^*} \right) \delta x dt$$

$$\Rightarrow \delta J = \int_{t_0}^{t_f} \left( \left. \frac{\partial L}{\partial x} \right|_{x^*} - \frac{d}{dt} \left. \frac{\partial L}{\partial \dot{x}} \right|_{\dot{x}^*} \right) \delta x dt + \left. \left( \left. \frac{\partial L}{\partial \dot{x}} \right|_{\dot{x}^*} \delta x \right) \right|_{t_0}^{t_f}$$

$$\delta J = 0, \text{由变分定理, 有} \begin{cases} \frac{\partial L}{\partial x} - \frac{d}{dt} \frac{\partial L}{\partial \dot{x}} = 0 \\ \left. \frac{\partial L}{\partial \dot{x}} \right|_{\dot{x}^*} \delta x(t_f) = \left. \frac{\partial L}{\partial \dot{x}} \right|_{\dot{x}^*} \delta x(t_0) \end{cases}$$

例1.  $J(x) = \int_0^{\frac{\pi}{2}} (\dot{x}^2(t) - x^2(t)) dt, x(0) = 0, x(\frac{\pi}{2}) = 2$ , 求  $J$  最小时的  $x^*$ .

解  $L(x, \dot{x}, t) = \dot{x}^2 - x^2$

由 Euler 方程  $\ddot{x} + x = 0 \Rightarrow x^* = 2 \sin t$

有约束泛函极值问题  $\begin{cases} \min_x J(x) = \int_{t_0}^{t_f} g(x, \dot{x}, t) dt \\ f(x, \dot{x}, t) = 0 \end{cases}$

取极值的必要条件:  $\begin{cases} \frac{\partial L}{\partial x} - \frac{d}{dt} \frac{\partial L}{\partial \dot{x}} = 0 \\ L(x, \dot{x}, \lambda, t) = g(x, \dot{x}, t) + \lambda(t) f(x, \dot{x}, t). \end{cases}$

例2. 对系统  $\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} u$

取  $J = \frac{1}{2} \int_0^2 u^2(t) dt, x(0) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, x(2) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ , 求  $J$  取极值的  $x^*(t)$  与  $u^*(t)$ .

解

$$g = \frac{1}{2} u^2$$

$$f = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} u - \begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix}$$

$$L(x, u, \lambda, t) = \frac{1}{2} u^2 + \lambda_1 (x_2 - \dot{x}_1) + \lambda_2 (u - \dot{x}_2)$$

$$\Rightarrow \begin{cases} \frac{\partial L}{\partial x_1} - \frac{d}{dt} \frac{\partial L}{\partial \dot{x}_1} = \lambda_1 = 0 \\ \frac{\partial L}{\partial x_2} - \frac{d}{dt} \frac{\partial L}{\partial \dot{x}_2} = \lambda_1 + \lambda_2 = 0 \\ \frac{\partial L}{\partial u} - \frac{d}{dt} \frac{\partial L}{\partial \dot{u}} = u + \lambda_2 = 0 \end{cases} \Rightarrow \begin{cases} \lambda_1 = a \\ \lambda_2 = -at + b \\ u = at - b \end{cases}$$

$$\Rightarrow \begin{cases} x_2 = \frac{1}{2} at^2 - bt + c \\ x_1 = \frac{1}{6} at^3 - \frac{1}{2} bt^2 + ct + d \end{cases} \Rightarrow \begin{cases} x_1^* = \frac{1}{2} t^3 - \frac{7}{4} t^2 + t + 1 \\ x_2^* = \frac{3}{2} t^2 - \frac{7}{2} t + 1 \end{cases}, u^* = 3t - 3.5$$

• J 极小值的充分条件 — Legendre 条件

$$ii) \begin{pmatrix} \frac{\partial^2 L}{\partial \dot{x}^2} & \frac{\partial^2 L}{\partial x \partial \dot{x}} \\ \frac{\partial^2 L}{\partial x \partial \dot{x}} & \frac{\partial^2 L}{\partial x^2} \end{pmatrix} \text{正定};$$

$$iii) \frac{\partial^2 L}{\partial \dot{x}^2} - \frac{d}{dt} \frac{\partial^2 L}{\partial x \partial \dot{x}} \geq 0, \quad \frac{\partial^2 L}{\partial \dot{x}^2} > 0$$

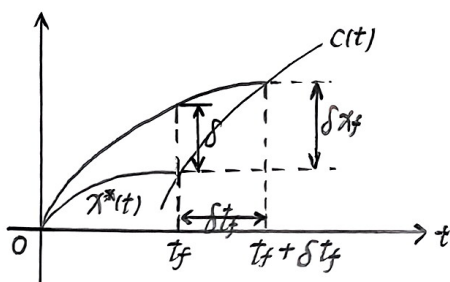
$$iii) \frac{\partial^2 L}{\partial \dot{x}^2} - \frac{d}{dt} \frac{\partial^2 L}{\partial x \partial \dot{x}} > 0, \quad \frac{\partial^2 L}{\partial \dot{x}^2} \geq 0$$

## 横截条件

末端时刻固定的横截条件.

|           |                                                                                                  |
|-----------|--------------------------------------------------------------------------------------------------|
| 固定起点和终点   | $x(t_0) = x_0, x(t_f) = x_f$                                                                     |
| 固定起点和自由终点 | $x(t_0) = x_0, \frac{\partial L}{\partial \dot{x}} _{t_f} = 0$                                   |
| 自由起点和固定终点 | $\frac{\partial L}{\partial \dot{x}} _{t_0} = 0, x(t_f) = x_f$                                   |
| 自由起点和自由终点 | $\frac{\partial L}{\partial \dot{x}} _{t_0} = 0, \frac{\partial L}{\partial \dot{x}} _{t_f} = 0$ |

末端时刻自由的横截条件



• ① 末端有约束时:

$$\begin{cases} \delta x(t_f) = \delta x_f - \dot{x}(t_f) \delta t_f \\ \delta x_f = \dot{C}(t_f) \delta t_f \end{cases}$$

② 末端无约束时:

$$\delta x(t_f) = \delta x_f - \dot{x}(t_f) \delta t_f.$$

• 末端时刻自由的横截条件:

$$\frac{\partial L}{\partial \dot{x}}|_{t_f} \delta x(t_f) - \frac{\partial L}{\partial \dot{x}}|_{t_0} \delta x(t_0) + L(x^*, \dot{x}^*, t)|_{t_f} \delta t_f = 0$$

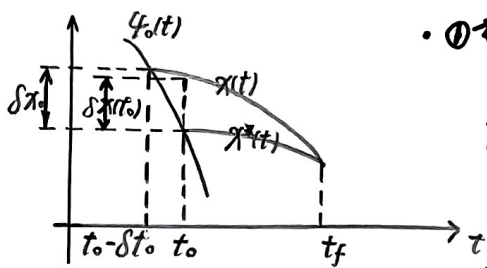
① 末端自由

$$\begin{cases} (L - \dot{x}(t) \frac{\partial L}{\partial \dot{x}})|_{t_f} = 0 \\ (\frac{\partial L}{\partial \dot{x}})|_{t_f} = 0 \\ x(t_0) = x_0 \end{cases}$$

② 末端约束 C(t)

$$\begin{cases} (L + (\dot{C} - \dot{x}) \frac{\partial L}{\partial \dot{x}})|_{t_f} = 0 \\ x(t_f) = C(t_f) \\ x(t_0) = x_0 \end{cases}$$

首端时刻自由的横截条件.



①首端有约束时:

$$\begin{cases} \delta x(t_0) = \delta x_0 + \dot{x}(t_0) \delta t_0 \\ \delta x_0 = -\dot{\varphi}_0(t_0) \delta t_0 \end{cases}$$

②首端无约束时:

$$\delta x(t_0) = \delta x_0 + \dot{x}(t_0) \delta t_0$$

·首端时刻自由的横截条件:

$$\left( \frac{\partial L}{\partial \dot{x}} \right)_{t_f} \delta x(t_f) - \left( \frac{\partial L}{\partial \dot{x}} \right)_{t_0} \delta x(t_0) + L(x^*, \dot{x}^*, t) \delta t_0 = 0$$

①首端自由

$$\begin{cases} (L - \dot{x} \frac{\partial L}{\partial \dot{x}})_{t_0} = 0 \\ \left( \frac{\partial L}{\partial \dot{x}} \right)_{t_0} = 0 \\ x(t_f) = x_f \end{cases}$$

②首端受约束

$$\begin{cases} (L - (\dot{\varphi}_0 - \dot{x}) \frac{\partial L}{\partial \dot{x}})_{t_0} = 0 \\ x(t_0) = \varphi_0(t_0) \\ x(t_f) = x_f \end{cases}$$

例3.  $t_0 = 0, x(0) = 1, x(t_f) = 2 - t_f, J = \int_0^{t_f} (1 + \dot{x}^2)^{\frac{1}{2}} dt$ , 求  $x^*(t)$  及  $t_f^*$

解

$$L(x, \dot{x}, t) = (1 + \dot{x}^2)^{\frac{1}{2}}$$

$$\Rightarrow \frac{d}{dt} \left( \frac{\dot{x}}{\sqrt{1 + \dot{x}^2}} \right) = 0 \Rightarrow x(t) = at + b$$

$$\text{由横截条件, } a = 1, b = 1 \Rightarrow x^*(t) = t + 1$$

$$x(t_f) = c(t_f) \Rightarrow t_f^* = \frac{1}{2}$$



### 3. 变分法解最优控制

· 最优控制问题  $\dot{x}(t) = f(x(t), u(t), t)$

$$x(t_0) = x_0$$

$$\text{性能泛函 } J(x) = \varphi[x(t_f), t_f] + \int_{t_0}^{t_f} L(x, u, t) dt$$

$$\text{目标集 } \varphi(x(t_f), t_f) = 0$$

o 简单的方法是解出  $u$ , 用前述 Euler 方程和横截条件求  $u^*$ ,  $x^*$ .

但  $u$  通常难以求出.

#### 末端时刻固定的解

$$\begin{cases} \dot{x}(t) = f(x(t), u(t), t) \\ x(t_0) = x_0 \\ \min_x J = \varphi[x(t_f)] + \int_{t_0}^{t_f} L(x, u, t) dt \\ \varphi[x(t_f)] = 0 \end{cases}$$

#### 最优解必要条件

##### □ 末端状态受约束

i) 构造哈密顿函数:

$$H(x, u, \lambda, t) = L(x, u, t) + \lambda^T(t) f(x, u, t)$$

满足正则方程:

$$\begin{cases} \dot{x}(t) = \frac{\partial H}{\partial \lambda} \\ \dot{\lambda}(t) = -\frac{\partial H}{\partial x} \end{cases} \quad (\text{协态方程})$$

iii) 满足边界条件

$$\begin{cases} x(t_0) = x_0 \\ \lambda(t_f) = \frac{\partial \varphi}{\partial x(t_f)} + \frac{\partial \varphi}{\partial x(t_f)} \gamma(t_f) \\ \varphi(x(t_f)) = 0 \end{cases}$$

iii) 满足极值条件:  $\frac{\partial H}{\partial u} = 0$

##### □ 末端自由

i) 构造哈密顿函数

$$H(x, u, \lambda, t) = L(x, u, t) + \lambda^T(t) f(x, u, t)$$

满足正则方程

$$\begin{cases} \dot{x}(t) = \frac{\partial H}{\partial \lambda} \\ \dot{\lambda}(t) = -\frac{\partial H}{\partial x} \end{cases}$$

iii) 满足边界条件

$$\begin{cases} x(t_0) = x_0 \\ \lambda(t_f) = \frac{\partial \varphi}{\partial x(t_f)} \end{cases}$$

iii) 满足极值条件:  $\frac{\partial H}{\partial u} = 0$

### ③ 末端状态固定:

若系统完全可控, 则

i) 构造哈密顿函数:

$$H(x, u, \lambda, t) = L(x, u, t) + \lambda(t)f(x, u, t)$$

满足正则方程

$$\begin{cases} \dot{x}(t) = \frac{\partial H}{\partial \lambda} \\ \dot{\lambda}(t) = -\frac{\partial H}{\partial x} \end{cases}$$

ii) 满足边界条件:

$$\begin{cases} x_0 = x(0) \\ x_f = x(t_f) \end{cases}$$

iii) 满足极值条件:  $\frac{\partial H}{\partial u} = 0$

例1. 
$$\begin{cases} \dot{x}(t) = u(t) \\ x(t_0) = x_0 \\ J = \frac{1}{2}cx^2(t_f) + \frac{1}{2}\int_{t_0}^{t_f} u^2(t)dt \end{cases}$$

$t_0, t_f$  给定,  $c > 0$ ,  $x(t_f)$  自由, 求  $J$  最小时的  $u^*(t)$  和  $J^*$ .

解  $f = u$ ,  $L = \frac{1}{2}u^2$ ,  $\varphi = \frac{1}{2}cx^2(t_f)$

$$H(x, u, \lambda, t) = \frac{1}{2}u^2 + \lambda u$$

$$\begin{cases} u + \lambda = 0 \Rightarrow u = -\lambda = -cx(t_f) \\ u = \dot{x} \\ \dot{\lambda} = 0 \Rightarrow \lambda \text{ 为常数} \\ \lambda(t_f) = \frac{\partial \varphi}{\partial x(t_f)} = cx(t_f) \end{cases}$$

$$\Rightarrow \dot{x} = -cx(t_f) \Rightarrow x = x_0 - cx(t_f)(t - t_0)$$

$$x^*(t_f) = \frac{x_0}{1 + c(t_f - t_0)} \Rightarrow u^*(t) = -\frac{cx_0}{1 + c(t_f - t_0)}$$

$$J^* = \frac{1}{2} \frac{cx_0^2}{1 + c(t_f - t_0)}$$

例2 
$$\begin{cases} \dot{x}_1(t) = x_2(t) \\ \dot{x}_2(t) = u(t) \\ x_1(0) = 0, x_2(0) = 0 \\ J = \frac{1}{2} \int_0^1 u^2 dt \end{cases}$$

$t_f = 1$ , 此时  $x_1(1) + x_2(1) = 1$ , 求  $u^*$

解  $\varphi(x(t_f)) = 0$ ,  $L = \frac{1}{2}u^2$ ,  $f_1 = x_2(t)$ ,  $f_2 = u(t)$ ,  $\varphi = x_1(1) + x_2(1) - 1$

$$H = \frac{1}{2}u^2 + \lambda_1 x_2 + \lambda_2 u$$

$$\begin{cases} \dot{\lambda}_1 = -\frac{\partial H}{\partial x_1} = 0 \\ \dot{\lambda}_2 = -\frac{\partial H}{\partial x_2} = -\lambda_1 \\ x_1(0) = x_2(0) = 0 \\ \lambda_1(1) = \frac{\partial \varphi}{\partial x_1(1)} \gamma = \gamma \\ \lambda_2(1) = \frac{\partial \varphi}{\partial x_2(1)} \gamma = \gamma \end{cases} \Rightarrow \begin{cases} \lambda_1 = C_1 \\ \lambda_2 = -C_1 t + C_2 \\ \dot{u} = C_1 t - C_2 \end{cases} \Rightarrow u^*(t) = -\frac{2}{7}(t-2).$$

$$\frac{\partial H}{\partial u} = u + \lambda_2 = 0$$

最优解充要条件

$$\begin{cases} \Theta(x(t_f), \gamma) = \varphi(x(t_f)) + \gamma(t) \varphi(x(t_f)) \\ H(x, u, \lambda, t) = L(x, u, t) + \lambda f(x, u, t) \end{cases}$$

充要条件为:

$$i) \begin{pmatrix} \frac{\partial^2 H}{\partial x^2} & \frac{\partial^2 H}{\partial u \partial x} \\ \frac{\partial^2 H}{\partial u \partial x} & \frac{\partial^2 H}{\partial u^2} \end{pmatrix} \text{正定}, \frac{\partial^2 \Theta}{\partial x(t_f)^2} \geq 0$$

$$ii) \begin{pmatrix} \frac{\partial^2 H}{\partial x^2} & \frac{\partial^2 H}{\partial u \partial x} \\ \frac{\partial^2 H}{\partial u \partial x} & \frac{\partial^2 H}{\partial u^2} \end{pmatrix} \text{半正定}, \frac{\partial^2 \Theta}{\partial x(t_f)^2} > 0.$$

末端时刻自由的解

$$\begin{cases} \dot{x}(t) = f(x(t), u(t), t) \\ x(t_0) = x_0 \\ \min_x J = \varphi(x(t_f), t_f) + \int_{t_0}^{t_f} L(x, u, t) dt \\ \varphi(x(t_f), t_f) = 0 \end{cases}$$

□ 末端状态受约束:

$$i) H(x, u, \lambda, t) = L(x, u, t) + \lambda(t) f(x, u, t)$$

↓

$$\begin{cases} \dot{x} = \frac{\partial H}{\partial \lambda} \\ \dot{\lambda} = -\frac{\partial H}{\partial x} \end{cases}$$

$$\text{ii)} \begin{cases} x(t_0) = x_0 \\ \lambda(t_f) = \frac{\partial \varphi}{\partial x(t_f)} + \frac{\partial \varphi}{\partial x(t_f)} x(t_f) \\ \varphi(x(t_f), t_f) = 0 \end{cases}$$

$$\text{iii)} \frac{\partial H}{\partial u} = 0$$

$$\text{iv)} H(t_f) = -\frac{\partial \varphi}{\partial t_f} - x(t_f) \frac{\partial \varphi}{\partial t_f}$$

## ② 末端自由

$$\text{ii)} H(x, u, \lambda, t) = L(x, u, t) + \lambda(t) f(x, u, t)$$

↓

$$\begin{cases} \dot{x} = \frac{\partial H}{\partial \lambda} \\ \dot{\lambda} = -\frac{\partial H}{\partial x} \end{cases}$$

$$\text{iii)} \begin{cases} x(t_0) = x_0 \\ \lambda(t_f) = \frac{\partial \varphi}{\partial x(t_f)} \end{cases}$$

$$\text{iii)} \frac{\partial H}{\partial u} = 0$$

$$\text{iv)} H(t_f) = -\frac{\partial \varphi}{\partial t_f}$$

## ③ 末端固定

$$\text{ii)} H(x, u, \lambda, t) = L(x, u, t) + \lambda(t) f(x, u, t)$$

↓

$$\begin{cases} \dot{x} = \frac{\partial H}{\partial \lambda} \\ \dot{\lambda} = -\frac{\partial H}{\partial x} \end{cases}$$

$$\text{iii)} \begin{cases} x(t_0) = x_0 \\ x(t_f) = x_f \end{cases}$$

$$\text{iii)} \frac{\partial H}{\partial u} = 0$$

$$\text{iv)} H(t_f) = -\frac{\partial \varphi}{\partial t_f}$$

角点条件与内点约束  $\rightarrow x(t)$  连续而不可微.

角点处的极值条件 对分段连续函数  $J(x)$ :

① 满足 Euler 方程;

$$\text{②} \begin{cases} \frac{\partial L}{\partial \dot{x}}|_{t_i^-} = \frac{\partial L}{\partial \dot{x}}|_{t_i^+} \\ (L - \dot{x} \frac{\partial L}{\partial \dot{x}})|_{t_i^-} = -(L - \dot{x} \frac{\partial L}{\partial \dot{x}})|_{t_i^+} \end{cases}$$

可取有极值.



内点约束下的最优控制

$$\begin{cases} \min_u J = \varphi(x(t_f)) + \int_{t_0}^{t_f} L(x, u, t) dt \\ \dot{x} = f(x, u, t) \\ x(0) = x_0 \\ n(x(t_i), t_i) = 0 \text{ (内点约束)} \end{cases}$$

• 最优解必要条件:

① 正则方程: 
$$\begin{cases} \dot{x} = \frac{\partial H}{\partial \lambda} \\ \dot{\lambda} = -\frac{\partial H}{\partial x} \end{cases}$$

② 横截条件 
$$\begin{cases} x(t_0) = x_0 \\ \lambda(t_f) = \frac{\partial \varphi}{\partial x(t_f)} \end{cases}$$

③ 极值条件 
$$\frac{\partial H}{\partial u} = 0$$

④ 内点约束 
$$\begin{cases} n(x(t_i), t_i) = 0 \\ \lambda(t_i^-) = \lambda(t_i^+) + \frac{\partial n}{\partial x(t_i)} \pi \\ H(t_i^-) = H(t_i^+) - \pi \frac{\partial n}{\partial t_i} \end{cases}$$

# 第三章 极小值原理

## 1. 连续系统的极小值原理

### 极小值原理 控制向量 $u \in \Omega$ (受约束)

表 3-1 定常系统极小值原理

| 末端时刻                                                | 性能指标                                                | 末端状态                              | 正则方程                                                                                                                                                 | 极小值条件                                                                     | 边界条件                                                                                                                                                 | $H$ 变化律                                                                                                                                              |                                                                           |                                                                             |
|-----------------------------------------------------|-----------------------------------------------------|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| $t_f$ 固定                                            | $J = \varphi[x(t_f)]$                               | 末端自由                              | $\dot{x} = \frac{\partial H}{\partial \lambda}$<br>$\dot{\lambda} = -\frac{\partial H}{\partial x}$                                                  | $H^* = H(x^*, \lambda^*, u^*) = \min_{u \in \Omega} H(x^*, \lambda^*, u)$ | $x(t_0) = x_0$<br>$\lambda(t_f) = \frac{\partial \varphi}{\partial x(t_f)}$                                                                          | $H^*(t) = H^*(t_f)$<br>$= \text{const}$                                                                                                              |                                                                           |                                                                             |
|                                                     |                                                     | 末端约束                              | 式中<br>$H = \lambda^T(t)f(x, u)$                                                                                                                      |                                                                           | $x(t_0) = x_0, \psi[x(t_f)] = 0$<br>$\lambda(t_f) = \left[ \frac{\partial \varphi}{\partial x} + \frac{\partial \psi^T}{\partial x} v \right]_{t_f}$ | $H^*(t) = H^*(t_f)$<br>$= \text{const}$                                                                                                              |                                                                           |                                                                             |
|                                                     | $J = \int_{t_0}^{t_f} L(x, u) dt$                   | 末端固定                              | $\dot{x} = \frac{\partial H}{\partial \lambda}$<br>$\dot{\lambda} = -\frac{\partial H}{\partial x}$<br>式中<br>$H = L(x, u) + \lambda^T(t)f(x, u)$     |                                                                           | $x(t_0) = x_0, x(t_f) = x_f$<br>( $\lambda(t_f)$ 未知)                                                                                                 | $H^*(t) = H^*(t_f)$<br>$= \text{const}$                                                                                                              |                                                                           |                                                                             |
|                                                     |                                                     | 末端自由                              |                                                                                                                                                      |                                                                           | $x(t_0) = x_0$<br>$\lambda(t_f) = 0$                                                                                                                 |                                                                                                                                                      |                                                                           |                                                                             |
|                                                     |                                                     | 末端约束                              |                                                                                                                                                      |                                                                           | $x(t_0) = x_0, \psi[x(t_f)] = 0$<br>$\lambda(t_f) = \left[ \frac{\partial \psi^T}{\partial x} v \right]_{t_f}$                                       |                                                                                                                                                      |                                                                           |                                                                             |
|                                                     | $J = \varphi[x(t_f)] + \int_{t_0}^{t_f} L(x, u) dt$ | 末端自由                              |                                                                                                                                                      |                                                                           | $x(t_0) = x_0$<br>$\lambda(t_f) = \frac{\partial \varphi}{\partial x(t_f)}$                                                                          | $H^*(t) = H^*(t_f)$<br>$= \text{const}$<br>(哈密顿函数为常数)                                                                                                |                                                                           |                                                                             |
|                                                     |                                                     | 末端约束                              |                                                                                                                                                      |                                                                           | $x(t_0) = x_0, \psi[x(t_f)] = 0$<br>$\lambda(t_f) = \left[ \frac{\partial \varphi}{\partial x} + \frac{\partial \psi^T}{\partial x} v \right]_{t_f}$ |                                                                                                                                                      |                                                                           |                                                                             |
|                                                     | $t_f$ 自由                                            | $J = \varphi[x(t_f)]$             |                                                                                                                                                      |                                                                           | 末端自由                                                                                                                                                 | $\dot{x} = \frac{\partial H}{\partial \lambda}$<br>$\dot{\lambda} = -\frac{\partial H}{\partial x}$                                                  | $H^* = H(x^*, \lambda^*, u^*) = \min_{u \in \Omega} H(x^*, \lambda^*, u)$ | $x(t_0) = x_0$<br>$\lambda(t_f) = \frac{\partial \varphi}{\partial x(t_f)}$ |
|                                                     |                                                     |                                   | 末端约束                                                                                                                                                 |                                                                           | 式中<br>$H = \lambda^T(t)f(x, u)$                                                                                                                      | $x(t_0) = x_0, \psi[x(t_f)] = 0$<br>$\lambda(t_f) = \left[ \frac{\partial \varphi}{\partial x} + \frac{\partial \psi^T}{\partial x} v \right]_{t_f}$ |                                                                           | $H^*(t_f) = 0$                                                              |
|                                                     |                                                     | $J = \int_{t_0}^{t_f} L(x, u) dt$ | 末端固定                                                                                                                                                 |                                                                           | $\dot{x} = \frac{\partial H}{\partial \lambda}$<br>$\dot{\lambda} = -\frac{\partial H}{\partial x}$<br>式中<br>$H = L(x, u) + \lambda^T(t)f(x, u)$     | $x(t_0) = x_0, x(t_f) = x_f$<br>( $\lambda(t_f)$ 未知)                                                                                                 |                                                                           | $H^*(t_f) = 0$                                                              |
| 末端自由                                                |                                                     |                                   | $x(t_0) = x_0$<br>$\lambda(t_f) = 0$                                                                                                                 |                                                                           |                                                                                                                                                      |                                                                                                                                                      |                                                                           |                                                                             |
| 末端约束                                                |                                                     |                                   | $x(t_0) = x_0, \psi[x(t_f)] = 0$<br>$\lambda(t_f) = \left[ \frac{\partial \psi^T}{\partial x} v \right]_{t_f}$                                       |                                                                           |                                                                                                                                                      |                                                                                                                                                      |                                                                           |                                                                             |
| $J = \varphi[x(t_f)] + \int_{t_0}^{t_f} L(x, u) dt$ |                                                     | 末端自由                              | $x(t_0) = x_0$<br>$\lambda(t_f) = \frac{\partial \varphi}{\partial x(t_f)}$                                                                          | $H^*(t_f) = 0$                                                            |                                                                                                                                                      |                                                                                                                                                      |                                                                           |                                                                             |
|                                                     |                                                     | 末端约束                              | $x(t_0) = x_0, \psi[x(t_f)] = 0$<br>$\lambda(t_f) = \left[ \frac{\partial \varphi}{\partial x} + \frac{\partial \psi^T}{\partial x} v \right]_{t_f}$ |                                                                           |                                                                                                                                                      |                                                                                                                                                      |                                                                           |                                                                             |



表 3-2 时变系统极小值原理

| 末端时刻                                                        | 性能指标                                                        | 末端状态                       | 正则方程                                                                                                                          | 极小值条件                                                                                                                                                     | 边界条件                                                                                                                                                      | H 变化律                                                                              |                                                                             |
|-------------------------------------------------------------|-------------------------------------------------------------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| $t_f$ 固定                                                    | $J = \varphi[x(t_f), t_f]$                                  | 末端自由                       | $\dot{x} = \frac{\partial H}{\partial \lambda}$<br>$\dot{\lambda} = -\frac{\partial H}{\partial x}$                           | $H^* = H(x^*, \lambda^*, u^*, t) = \min_{u \in U} H(x^*, \lambda^*, u, t)$                                                                                | $x(t_0) = x_0$<br>$\lambda(t_f) = \frac{\partial \varphi}{\partial x(t_f)}$                                                                               | $H^*(t)$<br>$= H^*(t_f)$<br>$-\int_t^{t_f} \frac{\partial H}{\partial \tau} d\tau$ |                                                                             |
|                                                             |                                                             | 末端约束                       | 式中<br>$H = \lambda^T(t)f(x, u, t)$                                                                                            |                                                                                                                                                           | $x(t_0) = x_0, \psi[x(t_f), t_f] = 0$<br>$\lambda(t_f) = \left[ \frac{\partial \varphi}{\partial x} + \frac{\partial \psi^T}{\partial x} v \right]_{t_f}$ |                                                                                    |                                                                             |
|                                                             | $J = \int_{t_0}^{t_f} L(x, u, t) dt$                        | 末端固定                       | $\dot{x} = \frac{\partial H}{\partial \lambda}$<br>$\dot{\lambda} = -\frac{\partial H}{\partial x}$<br>式中<br>$H = L(x, u, t)$ |                                                                                                                                                           | $x(t_0) = x_0, x(t_f) = x_f$<br>( $\lambda(t_f)$ 未知)                                                                                                      | $H^*(t)$<br>$= H^*(t_f)$<br>$-\int_t^{t_f} \frac{\partial H}{\partial \tau} d\tau$ |                                                                             |
|                                                             |                                                             | 末端自由                       |                                                                                                                               |                                                                                                                                                           | $x(t_0) = x_0$<br>$\lambda(t_f) = 0$                                                                                                                      |                                                                                    |                                                                             |
|                                                             |                                                             | 末端约束                       |                                                                                                                               |                                                                                                                                                           | $x(t_0) = x_0, \psi[x(t_f), t_f] = 0$<br>$\lambda(t_f) = \frac{\partial \psi^T}{\partial x(t_f)} v$                                                       |                                                                                    |                                                                             |
|                                                             | $J = \varphi[x(t_f), t_f] + \int_{t_0}^{t_f} L(x, u, t) dt$ | 末端自由                       | $H = L(x, u, t) + \lambda^T(t)f(x, u, t)$                                                                                     |                                                                                                                                                           | $x(t_0) = x_0$<br>$\lambda(t_f) = \frac{\partial \varphi}{\partial x(t_f)}$                                                                               | $H^*(t)$<br>$= H^*(t_f)$<br>$-\int_t^{t_f} \frac{\partial H}{\partial \tau} d\tau$ |                                                                             |
|                                                             |                                                             | 末端约束                       |                                                                                                                               |                                                                                                                                                           | $x(t_0) = x_0, \psi[x(t_f), t_f] = 0$<br>$\lambda(t_f) = \left[ \frac{\partial \varphi}{\partial x} + \frac{\partial \psi^T}{\partial x} v \right]_{t_f}$ |                                                                                    |                                                                             |
|                                                             | $t_f$ 自由                                                    | $J = \varphi[x(t_f), t_f]$ | 末端自由                                                                                                                          |                                                                                                                                                           | $\dot{x} = \frac{\partial H}{\partial \lambda}$<br>$\dot{\lambda} = -\frac{\partial H}{\partial x}$                                                       | $H^* = H(x^*, \lambda^*, u^*, t) = \min_{u \in U} H(x^*, \lambda^*, u, t)$         | $x(t_0) = x_0$<br>$\lambda(t_f) = \frac{\partial \varphi}{\partial x(t_f)}$ |
| 末端约束                                                        |                                                             |                            | 式中<br>$H = \lambda^T(t)f(x, u, t)$                                                                                            | $x(t_0) = x_0, \psi[x(t_f), t_f] = 0$<br>$\lambda(t_f) = \left[ \frac{\partial \varphi}{\partial x} + \frac{\partial \psi^T}{\partial x} v \right]_{t_f}$ | $H^*(t_f^*) =$<br>$-\left[ \frac{\partial \varphi}{\partial t} + v^T \frac{\partial \psi}{\partial t} \right]_{t_f}$                                      |                                                                                    |                                                                             |
| $J = \int_{t_0}^{t_f} L(x, u, t) dt$                        |                                                             | 末端固定                       | $\dot{x} = \frac{\partial H}{\partial \lambda}$<br>$\dot{\lambda} = -\frac{\partial H}{\partial x}$<br>式中<br>$H = L(x, u, t)$ | $x(t_0) = x_0, x(t_f) = x_f$<br>( $\lambda(t_f)$ 未知)                                                                                                      | $H^*(t_f^*) = 0$                                                                                                                                          |                                                                                    |                                                                             |
|                                                             |                                                             | 末端自由                       |                                                                                                                               | $x(t_0) = x_0$<br>$\lambda(t_f) = 0$                                                                                                                      | $H^*(t_f^*) = 0$                                                                                                                                          |                                                                                    |                                                                             |
|                                                             |                                                             | 末端约束                       |                                                                                                                               | $x(t_0) = x_0, \psi[x(t_f), t_f] = 0$<br>$\lambda(t_f) = \frac{\partial \psi^T}{\partial x(t_f)} v$                                                       | $H^*(t_f^*) =$<br>$= -v^T \frac{\partial \psi}{\partial t_f}$                                                                                             |                                                                                    |                                                                             |
| $J = \varphi[x(t_f), t_f] + \int_{t_0}^{t_f} L(x, u, t) dt$ |                                                             | 末端自由                       | $H = L(x, u, t) + \lambda^T(t)f(x, u, t)$                                                                                     | $x(t_0) = x_0$<br>$\lambda(t_f) = \frac{\partial \varphi}{\partial x(t_f)}$                                                                               | $H^*(t_f^*) = -\frac{\partial \varphi}{\partial t_f}$                                                                                                     |                                                                                    |                                                                             |
|                                                             |                                                             | 末端约束                       |                                                                                                                               | $x(t_0) = x_0, \psi[x(t_f), t_f] = 0$<br>$\lambda(t_f) = \left[ \frac{\partial \varphi}{\partial x} + \frac{\partial \psi^T}{\partial x} v \right]_{t_f}$ | $H^*(t_f^*) =$<br>$-\left[ \frac{\partial \varphi}{\partial t} + v^T \frac{\partial \psi}{\partial t} \right]_{t_f}$                                      |                                                                                    |                                                                             |

ii) 容许控制条件放宽, 经典变分法中  $u \in R^n$ .

iii) 最优控制使哈密顿函数全局最小.

iiii) 不要求  $H$  对控制的可微性.

例1.  $\begin{cases} \dot{x}_1(t) = -x_1(t) + u(t), & x_1(0) = 1 \\ \dot{x}_2(t) = x_1(t), & x_2(0) = 0 \end{cases}$

$|u(t)| \leq 1$ , 末端状态自由,  $J = x_2(1)$ , 求  $u^*$

解  $\varphi(x(t_f)) = x_2(1), t_f = 1$

$$H(u, x, \lambda) = \lambda_1(-x_1 + u) + \lambda_2 x_1$$

$$\begin{cases} \dot{\lambda}_1 = -\frac{\partial H}{\partial x_1} = \lambda_1 - \lambda_2 \\ \dot{\lambda}_2 = -\frac{\partial H}{\partial x_2} = 0 \end{cases}$$

$$\begin{cases} \lambda_1 = c_1 e^t + c_2 \\ \lambda_2 = c_2 \end{cases}$$

$$\begin{cases} \lambda_1(1) = \frac{\partial \varphi}{\partial x_1(1)} = 0 \\ \lambda_2(1) = \frac{\partial \varphi}{\partial x_2(1)} = 1 \end{cases} \Rightarrow \begin{cases} \lambda_1 = 1 - e^{t-1} \\ \lambda_2 = 1 \end{cases}$$

令  $H$  最小,  $u$  仅与  $\lambda_1$  有关, 令  $u = -\text{sgn}(\lambda_1(t)) = 0$

由  $\lambda_1(t) > 0$ , 则  $u^*(t) = -1$ .

例2. 系统方程  $\dot{x}(t) = x(t) - u(t), x(0) = 5$

$0.5 \leq u(t) \leq 1, J = \int_0^1 (x(t) + u(t)) dt$ , 求  $u^*$

解  $H(u, x, \lambda) = x + u + \lambda(x - u)$

令  $u(1-\lambda)$  最小, 则  $u^* = \begin{cases} 1 & \lambda > 1 \\ 0.5 & \lambda < 1 \end{cases}$

$$\dot{\lambda} = -\frac{\partial H}{\partial x} = -1 - \lambda \Rightarrow \lambda = ce^{-t} - 1$$

$$\lambda(1) = ce^{-1} - 1 = 0 \Rightarrow \lambda(t) = e^{1-t} - 1$$

$$\text{则 } u^* = \begin{cases} 1 & 0 \leq t < 0.307 \\ 0.5 & 0.307 \leq t \leq 1 \end{cases}$$

例3.  $\begin{cases} \dot{x}_1(t) = x_2(t) \\ \dot{x}_2(t) = u(t) \end{cases}$

$x_1(0) = x_2(0) = 0, x_1(t_f) = x_2(t_f) = \frac{1}{4}, |u(t)| \leq 1, J = \int_0^{t_f} u^2(t) dt$ , 求  $u^*$ .

解  $H(u, x, \lambda) = \lambda_1 x_2 + \lambda_2 u + u^2 = (u + \frac{1}{2}\lambda_2)^2 + \lambda_1 x_2 - \frac{1}{4}\lambda_2^2$

$$\text{则 } u^*(t) = \begin{cases} -1 & \lambda_2 > 2 \\ -\frac{1}{2}\lambda_2(t) & |\lambda_2| \leq 2 \\ 1 & \lambda_2 < -2 \end{cases}$$

$$\text{同时 } \begin{cases} u^{*2}(0) + \lambda_1(0)x_2^*(0) + \lambda_2(0)u^*(0) = 0 \\ u^{*2}(t_f) + \lambda_1(t_f)x_2^*(t_f) + \lambda_2(t_f)u^*(t_f) = 0 \end{cases} \Rightarrow u^*(0) = 0$$

$$\begin{cases} \dot{\lambda}_1 = -\frac{\partial H}{\partial x_1} = 0 \\ \dot{\lambda}_2 = -\frac{\partial H}{\partial x_2} = -\lambda_1 \end{cases} \Rightarrow \begin{cases} \lambda_1(t) = 2C_1 \\ \lambda_2(t) = -2C_1 t + C_2 \end{cases} \Rightarrow C_1 = \frac{1}{18}, t_f^* = 3, u^*(t) = \frac{1}{18}t$$

## 2. 离散系统极小值原理

### 离散 Euler 方程与横截条件

• 离散系统最优控制:

$$x(n+1) = f(x(n), u(n), n)$$

取性能离散泛函  $J = \sum_{n=0}^{N-1} L(x(n), u(n), n)$

① 性能离散泛函  $J$  的最优值必要条件:

ii) 满足离散 Euler 方程: (可用拉格朗日乘子将问题转化为无约束问题)

$$\begin{cases} \frac{\partial L_n}{\partial x(n)} + \frac{\partial L_{n-1}}{\partial x(n)} = 0 \\ \frac{\partial L_n}{\partial u(n)} = 0 \end{cases} \quad (L_n = L + \lambda(n+1)(f - x(n+1)))$$

iii) 满足横截条件

$$\left. \frac{\partial L_n}{\partial x(n)} \right|_N \delta x(N) - \left. \frac{\partial L_{n-1}}{\partial x(n)} \right|_0 \delta x(0) = 0$$

例1.

$$x(n+1) = x(n) + u(n)$$

$$x(0) = 1, x(5) = 0, J = \frac{1}{2} \sum_{n=0}^4 u^2(n), \text{ 求 } u^*(n).$$

解

$$\begin{cases} L_n = \frac{1}{2} u^2(n) + \lambda(n+1)(-x(n+1) + x(n) + u(n)) \\ L_{n-1} = \frac{1}{2} u^2(n-1) + \lambda(n)(-x(n) + x(n-1) + u(n-1)) \end{cases}$$

$$\frac{\partial L_n}{\partial x(n)} = \lambda(n+1) \quad \frac{\partial L_{n-1}}{\partial x(n)} = -\lambda(n), \quad \frac{\partial L_n}{\partial u(n)} = u(n) + \lambda(n+1)$$

由 Euler 方程

$$\begin{cases} \lambda(n+1) = \lambda(n) = c \\ u(n) = -\lambda(n+1) = -c \end{cases}$$

$$\Rightarrow x(n) = x(0) - nc \Rightarrow c = \frac{1}{5}$$

$$\text{则 } u^*(n) = -0.2$$

### 离散极小值原理

• 对离散系统  $x(n+1) = f(x(n), u(n), n), x(0) = x_0$

$$J = \varphi(x(N), N) + \sum_{n=0}^{N-1} L(x(n), u(n), n)$$

$u(n) \in \Omega$ ,  $\Omega$  为容许控制域; 末端状态约束为  $\varphi(x(N), N) = 0$ .

$u^*(n), x^*(n), \lambda(n), \gamma$  满足:

ii) 满足正则方程, 哈密顿函数  $H(n) = L(x(n), u(n), n) + \lambda(n+1)f(x(n), u(n), n)$

$$\begin{cases} x(n+1) = \frac{\partial H(n)}{\partial \lambda(n+1)} \\ \lambda(n) = \frac{\partial H(n)}{\partial x(n)} \end{cases}$$



iii) 满足边界条件:

$$\begin{cases} x(0) = x_0 \\ \psi(x(N), N) = 0 \\ \lambda(N) = \frac{\partial \varphi(x(N), N)}{\partial x(N)} + \gamma \frac{\partial \psi(x(N), N)}{\partial x(N)} \quad (\text{末端自由: } \lambda(N) = \frac{\partial \varphi(x(N), N)}{\partial x(N)}) \end{cases}$$

iii)  $H$  对  $u^*$  取极小值:

$$H(x^*(n), u^*(n), \lambda(n+1), n) = \min_{u \in \Omega} H(x^*(n), u^*(n), \lambda(n+1), n)$$

$$\text{若 } u \text{ 无约束: } \frac{\partial H(n)}{\partial u(n)} = 0$$

例2.  $x(n+1) = x(n) + u(n)$ ,  $x(0) = x_0$ ,  $J = \sum_{n=0}^2 (x^2(n) + u^2(n))$ , 求  $u^*$

解  $H(n) = x^2(n) + u^2(n) + \lambda(n+1)(x(n) + u(n))$

$$\begin{cases} \lambda(n) = \frac{\partial H(n)}{\partial x(n)} = 2x(n) + \lambda(n+1) \\ \frac{\partial H(n)}{\partial u(n)} = 2u(n) + \lambda(n+1) = 0 \\ \lambda(3) = 0 \end{cases} \Rightarrow \begin{aligned} u(n) &= -\frac{1}{2}\lambda(n+1) \\ \lambda(n+1) &= \lambda(n) - 2x(n) \end{aligned}$$

$$\Rightarrow \begin{cases} u^*(0) = -\frac{3}{5}x_0 \\ u^*(1) = -\frac{1}{5}x_0 \\ u^*(2) = 0 \end{cases}$$

### 连续系统的离散化法

- 采用连续系统离散化后求的最优控制, 在原连续问题不严格最优, 离散化后亦不严格最优 (仅在  $T \rightarrow 0$  时等效).

表 3-3 连续与离散极小值原理比较表

| 项 目              | 连续极小值原理                                                                                                                    | 离散极小值原理                                                                                                        |
|------------------|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| 系 统              | $\dot{x}(t) = f[x(t), u(t), t]$                                                                                            | $x(k+1) = f[x(k), u(k), k], \quad k=0, 1, \dots, N-1$                                                          |
| 初始条件             | $x(t_0) = x_0$                                                                                                             | $x(0) = x_0$                                                                                                   |
| 性能指标             | $J = \varphi[x(t_f), t_f] + \int_{t_0}^{t_f} L(x, u, t) dt$                                                                | $J = \varphi[x(N), N] + \sum_{k=0}^{N-1} l[x(k), u(k), k]$                                                     |
| 极值问题             | 求 $u^*(t)$ , 使 $J = \min$                                                                                                  | 求 $u^*(k)$ , 使 $J = \min, \quad k=0, 1, \dots, N-1$                                                            |
| 方法特点             | 引入协态向量 $\lambda(t)$                                                                                                        | 引入协态向量序列 $\lambda(k), \quad k=1, 2, \dots, N$                                                                  |
| 哈密顿函数            | $H(x, u, \lambda, t) = L(x, u, t) + \lambda^T f(x, u, t)$                                                                  | $H(k) = L[x(k), u(k), k] + \lambda^T(k+1) f[x(k), u(k), k]$<br>$k=0, 1, \dots, N-1$                            |
| 正则方程             | $\dot{x}(t) = \frac{\partial H}{\partial \lambda}, \quad \dot{\lambda}(t) = -\frac{\partial H}{\partial x}$                | $x(k+1) = \frac{\partial H(k)}{\partial \lambda(k+1)}, \quad \lambda(k) = \frac{\partial H(k)}{\partial x(k)}$ |
| 横截条件<br>(末端自由)   | $\lambda(t_f) = \frac{\partial \varphi[x(t_f), t_f]}{\partial x(t_f)}$<br>$\varphi[x(t_f), t_f] = 0$ 时, $\lambda(t_f) = 0$ | $\lambda(N) = \frac{\partial \varphi[x(N), N]}{\partial x(N)}$<br>$\varphi[x(N), N] = 0$ 时, $\lambda(N) = 0$   |
| 极值条件<br>(控制无约束)  | $\frac{\partial H}{\partial u} = 0$                                                                                        | $\frac{\partial H(k)}{\partial u(k)} = 0, \quad k=0, 1, \dots, N-1$                                            |
| 极小值条件<br>(控制有约束) | $H(x^*, u^*, \lambda, t) = \min_{u \in \Omega} H(x^*, u, \lambda, t)$                                                      | $H[x^*(k), u^*(k), \lambda(k+1), k]$<br>$= \min_{u(k) \in \Omega} H[x^*(k), u(k), \lambda(k+1), k]$            |

### 3. 时间最优控制

#### 时间最优控制

• 时间最优控制问题:

$$\min_{|u_j(t)| \leq 1} J = \int_{t_0}^{t_f} dt = t_f - t_0$$

$$\begin{cases} \dot{x}(t) = f(x(t), t) + B(x(t), t)u(t) \\ \varphi(x(t_f), t_f) = 0 \end{cases}$$

哈密顿函数  $H(x, \lambda, u, t) = 1 + \lambda(t)(f(x(t), t) + B(x(t), t)u(t))$

正则方程  $\dot{x} = f(x(t), t) + B(x(t), t)u(t)$

$$\dot{\lambda} = -\frac{\partial f}{\partial x} \lambda(t) - \frac{\partial (Bu)}{\partial x} \lambda$$

极小值条件:

$$\min \{1 + \lambda(t)f(x(t), t) + \lambda(t)B(x(t), t)u(t)\} \Rightarrow \min \{\lambda(t)B(x(t), t)u(t)\}$$

则令  $B(x, t) = (b_1(x, t), b_2(x, t), \dots, b_m(x, t))$

$$g_j(t) = b_j^T(x, t) \lambda(t)$$

$$u_j^*(t) = -\text{sgn}[g_j(t)]$$

○ 奇异问题和正常问题

ii 奇异问题: 若在  $[t_1, t_2]$  内存在

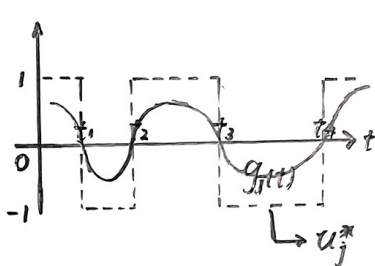
$$g_j(t) = b_j^T(x, t) \lambda(t) = 0$$

则时间最优控制奇异;

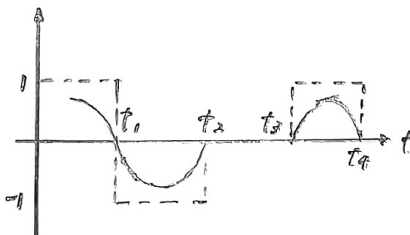
iii 正常问题: 若在  $[t_0, t_f]$  内仅有有限时间点使任一

$$g_j(t) = b_j^T(x, t) \lambda(t) = 0$$

则时间最优控制正常。



正常问题



奇异问题

**Bang-Bang 控制原理** 对于正常的时间控制问题, 时间最优控制:

$$u^*(t) = -\text{sgn}[B(x^*(t), t)\lambda(t)]$$

○ 若为奇异控制, 应另选最优控制律。

#### 线性定常系统的時間最优控制

• 时间最优控制问题:  $\dot{x} = Ax + Bu$

$$x(0) = x_0, \quad x(t_f) = 0, \quad J = \int_0^{t_f} dt, \quad |u_j(t)| \leq 1$$



o 奇异问题和正常问题  $B = (b, b_1, \dots, b_m)$

ii) 若  $m$  个矩阵  $G_j = (b_j, Ab_j, \dots, A^{n-1}b_j)$

有一个奇异, 则为奇异问题.

iii) 若  $G_j$  的秩为  $n$ , 则为正常问题.

iii) 系统正常  $\longleftrightarrow$  系统可控

(S1 系统则等价)

o 时间最优控制:

ii) 正则方程 
$$\begin{cases} \dot{x} = Ax + Bu \\ \dot{\lambda} = -A^T \lambda \end{cases}$$

iii) 边界条件  $x(0) = x_0, x(t_f) = 0$

iii)  $H$  变比律  $H^*(t_f) = 0$

iv) 极小值条件  $u_j^*(t) = -\text{sgn}(b_j^T \lambda) \begin{cases} +1 \\ -1 \end{cases}$

例1.  $\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} u(t), x_1(0) = x_2(0) = 1, x_1(t_f) = x_2(t_f) = 0, |u(t)| \leq 1$

求  $u^*$  使  $J = \int_0^{t_f} dt$  最小.

解 系统能控且正常, 则 Bang-Bang 控制为最优控制

$$H = \lambda_1 x_1 + \lambda_2 u + 1$$

$$u^* = \begin{cases} 1 & \lambda_2(t) < 0 \\ -1 & \lambda_2(t) > 0 \end{cases}$$

再由协态方程 
$$\begin{cases} \dot{\lambda}_1 = 0 \\ \dot{\lambda}_2 = -\lambda_1 \end{cases}$$

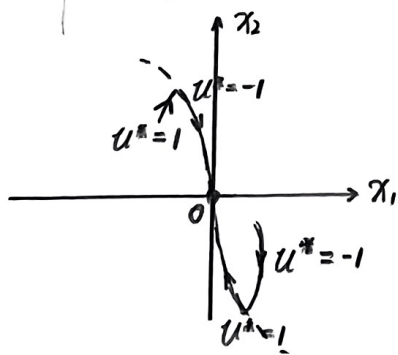
若  $u^*(t) = 1$ , 则

$$\begin{cases} x_2(t) = x_2(0) + t \\ x_1(t) = \frac{1}{2} t^2 + x_2(0)t + x_1(0) \end{cases} \Rightarrow x_1 = \frac{1}{2} x_2^2 (x_1(0) - \frac{1}{2} x_2^2(0))$$

满足末态的相轨迹为  $\gamma_+ = \{x_1 = \frac{1}{2} x_2^2, x_2 \leq 0\}$

若  $u^*(t) = -1$ , 则

$$\gamma_- = \{x_1 = -\frac{1}{2} x_2^2, x_2 \geq 0\}, \gamma_+ \cup \gamma_- \text{ 为开关曲线.}$$



现有  $x_1(0) = x_2(0) = 1$ , 则  $u^*(t)$  从  $-1 \rightarrow +1$

解方程可得

$$u^* = \begin{cases} -1 & 0 \leq t < 2.22 \\ 1 & 2.22 \leq t \leq 3.44 \end{cases}$$



注: 连续系统的时间最优控制常伴随轨线分析, 首先可得控制“序列”, 再根据相轨迹和控制得到最优控制轨线.

### 离散系统的时间最优控制

· 若  $u$  不受约束, 则任一  $x(0)$  可以在  $nT_s$  内到达原点;

若  $u$  受约束, 则  $N \geq n$  使  $x(N) = 0$ .

· 对  $x(n+1) = Ax(n) + Bu(n)$ ,  $x(0) = x_0$ ,  $A$  可逆, 系统能控, 则

$$x(0) = GU^*$$

$$G = (I - A^{-1} \quad -A^{-2} \quad \dots \quad -A^{-N})B$$

$$U^* = \begin{pmatrix} u^*(0) \\ u^*(1) \\ \vdots \\ u^*(N-1) \end{pmatrix}$$

可得  $U^*$  使  $x(N) = 0$ .

例2. 
$$\begin{pmatrix} x_1(n+1) \\ x_2(n+1) \end{pmatrix} = \begin{pmatrix} 1 & 1 - e^{-T_s} \\ e^{-T_s} & 1 - e^{-T_s} \end{pmatrix} \begin{pmatrix} x_1(n) \\ x_2(n) \end{pmatrix} + \begin{pmatrix} e^{-T_s} + T_s - 1 \\ 1 - e^{-T_s} \end{pmatrix} u(n)$$

求  $u^*(n)$  使  $x(2) = 0$ .

解 
$$G = (I - A^{-1}b \quad -A^{-2}b)$$

$$= - \begin{pmatrix} T - e^{-T} + 1 & T + e^T - e^{2T} \\ e^{-T} - 1 & e^{2T} - e^T \end{pmatrix} = (g_0 \quad g_1)$$

$$\begin{cases} x(0) = g_0 u(0) + g_1 u(1) \\ x(1) = g_0 u(1) \end{cases} \Rightarrow u^* = \frac{e^T - e^{2T}}{T(1 + e^{2T} - 2e^T)} x_1(n) + \frac{T + e^T - e^{2T}}{T(1 + e^{2T} - 2e^T)} x_2(n)$$

#### 4. 燃料最优控制

· 燃料最优控制:  $\min J = \int_0^{t_f} \Sigma c_j |u_j(t)| dt$ .

#### 二次积分模型的燃料控制

$$\begin{cases} \dot{x}_1(t) = x_2(t) \\ \dot{x}_2(t) = u \end{cases}$$

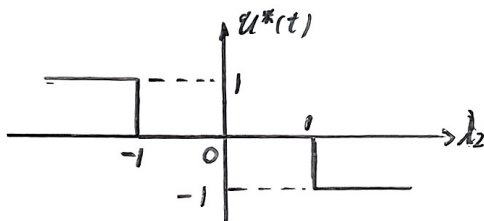
$|u(t)| \leq 1$ ,  $J = \int_0^{t_f} |u(t)| dt$ , 系统从任一初态转移至原点.

$$H(x, \lambda, u) = |u(t)| + \lambda_1(t) x_2(t) + \lambda_2(t) u(t)$$

$$u(t) \geq 0 \text{ 时, } \Rightarrow \begin{cases} u^*(t) = 0, & \lambda_2(t) > -1 \\ u^*(t) \in [0, 1], & \lambda_2(t) = -1 \\ u^*(t) = 1, & \lambda_2(t) < -1 \end{cases}$$

$$u(t) \leq 0 \text{ 时 } \Rightarrow \begin{cases} u^*(t) = 0, & \lambda_2(t) < 1 \\ u^*(t) \in [-1, 0], & \lambda_2(t) = -1 \\ u^*(t) = -1, & \lambda_2(t) > 1 \end{cases}$$

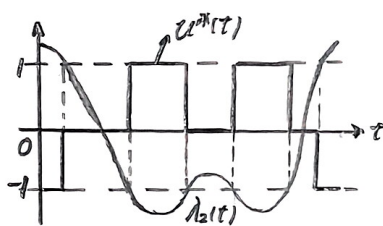
·  $u^*(t)$  与  $\lambda_2(t)$  有关, 呈死区特性



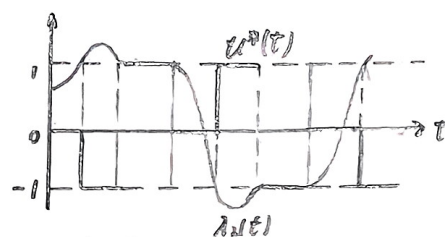
#### ① 正常与奇异情况

iii 在有限点上  $|\lambda_2(t)| = 1$ , 为正常情况;

iiii 存在  $[t_1, t_2]$  使  $|\lambda_2(t)| = 1$ , 为奇异情况;



正常情况



奇异情况

↓  
Bang-zero-Bang 控制

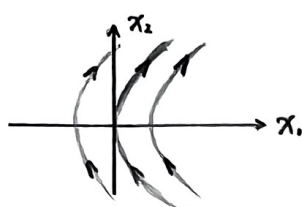
$$\textcircled{0} \begin{cases} \lambda_1 = c_1 \\ \lambda_2 = -c_1 t + c_2 \end{cases} \Rightarrow \begin{cases} c_1 = 0 \text{ 时奇异} \\ c_1 \neq 0 \text{ 时不奇异 (三位控制, 最多两次切换)} \end{cases}$$

↓

$$\begin{cases} -1 & ; & 0 & ; & 1 \\ -1, 0 & ; & 0, 1 & ; & 1, 0 & ; & 0, -1 \\ +1, 0, -1 & ; & -1, 0, +1. \end{cases}$$

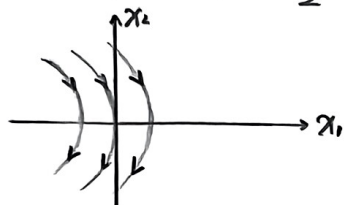
## 正常情况的最优控制

$u = 1$  时,  $\dot{x}_1 = \frac{1}{2}x_2^2 + C$ , 相轨迹为一族抛物线.



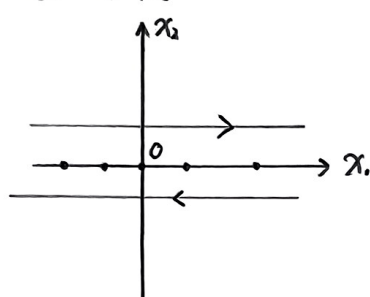
$\gamma_+ = \{x_1 = \frac{1}{2}x_2^2, x_2 \leq 0\}$  可达原点.

$u = -1$  时,  $\dot{x}_1 = -\frac{1}{2}x_2^2 + C$

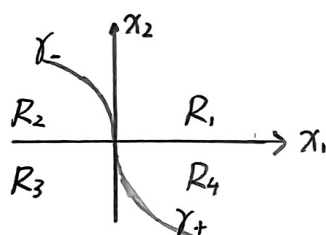


$\gamma_- = \{x_1 = -\frac{1}{2}x_2^2, x_2 \geq 0\}$  可达原点.

$u = 0$  时



则末态  $u=0$  的控制必不最优, 若末态  $u=0$ , 系统不可达状态空间原点.



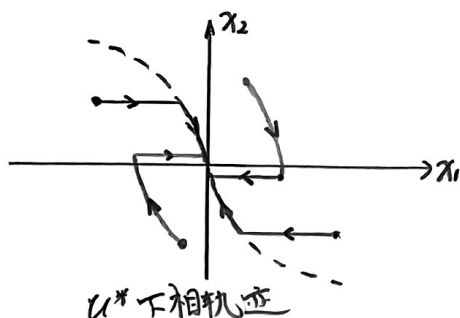
(i) 若  $(x_{10}, x_{20}) \in \gamma_+$ , 则  $u^* = 1$  为唯一燃料最优控制;

若  $(x_{10}, x_{20}) \in \gamma_-$ , 则  $u^* = -1$  为唯一燃料最优控制;

(ii) 若  $(x_{10}, x_{20}) \in R_4$ , 则  $u^* = \{0, 1\}$  为唯一燃料最优控制

$(x_{10}, x_{20}) \in R_2$ , 则  $u^* = \{0, -1\}$  为唯一燃料最优控制

(iii) 若  $(x_{10}, x_{20}) \in R_1 / R_3$ , 则  $u^*$  无解, 但有最接近于最小消耗的方法.



## 5. 时间-燃料最优控制

• 时间-燃料最优控制:  $\min J = \rho t_f + \int_0^{t_f} |u(t)| dt$

$\rho = 0 \Rightarrow$  燃料最优  $\rho = \infty \Rightarrow$  时间最优.

### 一维系统的时间-燃料控制

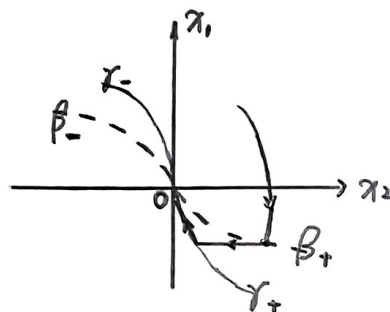
• 时间-燃料控制一定是正常情况.

$$u^* = \{1\}, \{-1\}, \{0, 1\}, \{0, -1\}, \{1, 0, -1\}, \{-1, 0, 1\}$$

• 系统存在两条切换线:

$$\gamma = \{x_1 = -\frac{1}{2}|x_2|x_2\}$$

$$\beta = \{x_1 = -\frac{\rho+4}{2\rho}|x_2|x_2\}$$



$\begin{cases} \rho \rightarrow \infty \text{ 时 } \beta_- \text{ 与 } \gamma_+ \text{ 重合, } \beta_+ \text{ 与 } \gamma_- \text{ 重合, 时间最优} \\ \rho = 0 \text{ 时, } \beta_-, \beta_+ \text{ 与 } x_1 \text{ 重合, 燃料最优.} \end{cases}$



## 第四章 线性二次型问题

### 1. 线性二次型问题概论

- 线性二次型问题：系统为线性系统，性能指标为  $u, x$  的二次型函数。
  - 线性最优控制结果可用于小信号非线性控制；
  - 线性二次型最优控制有良好频率响应。

### 线性二次型问题

线性系统状态方程 
$$\begin{cases} \dot{x}(t) = A(t)x(t) + B(t)u(t) \\ y(t) = C(t)x(t) \end{cases}$$

$u(t)$  无约束,  $y_r(t)$  为期望输出, 误差  $e(t) = y_r(t) - y(t)$

性能指标  $J = \frac{1}{2} e^T(t_f) F e(t_f) + \frac{1}{2} \int_{t_0}^{t_f} (e^T(t) Q(t) e(t) + u^T(t) R(t) u(t)) dt$

① 第一积分过程项  $0.5 \int_{t_0}^{t_f} e^T Q e dt \Rightarrow$  动态跟踪误差总度量  
 $Q$ : 对称时变矩阵

第二积分过程项  $0.5 \int_{t_0}^{t_f} u^T R u dt \Rightarrow$  控制能量消耗总度量  
 $R$ : 正定对称时变矩阵

终值项  $0.5 e^T(t_f) F e(t_f) \Rightarrow$  末态跟踪误差要求  
 $F$ : 半正定常数阵

### 线性二次型问题的类型

① 状态调节器  $C(t) = E, y_r(t) = 0$

$\Downarrow$   
 $J = \frac{1}{2} x^T(t_f) F x(t_f) + \frac{1}{2} \int_{t_0}^{t_f} (x^T Q x + u^T R u) dt$

系统偏离平衡状态时, 要求系统产生控制向量使  $x$  保持在平衡状态附近。

② 输出调节器  $y_r(t) = 0$

$\Downarrow$   
 $J = \frac{1}{2} y^T(t_f) F y(t_f) + \frac{1}{2} \int_{t_0}^{t_f} (y^T Q y + u^T R u) dt$

系统偏离平衡状态时, 要求系统产生控制向量使  $y$  保持在平衡状态附近。

③ 跟踪系统  $y_r(t) \neq 0$ 。

## 2. 状态调节器(LQR)

### 有限时间状态调节器

• 有限时间状态调节器问题:

$$\dot{x}(t) = A(t)x(t) + B(t)u(t)$$

$$x_0(t) = x_0$$

$u(t)$  无约束, 确定  $u^*$  使  $J$  最小.

$$J = \frac{1}{2}x^T(t_f)Fx(t_f) + \frac{1}{2}\int_{t_0}^{t_f} (x^T(t)Q(t)x(t) + u^T(t)R(t)u(t))dt$$

$$F = F^T \geq 0, \quad Q = Q^T \geq 0, \quad R = R^T > 0.$$

• 最优控制的充要必要条件:

$$u^*(t) = -R^{-1}(t)B^T(t)P(t)x(t)$$

其中,  $P(t)$  由黎卡提方程决定:

$$\begin{cases} -\dot{P}(t) = P(t)A(t) + A^T(t)P(t) - P(t)B(t)R^{-1}(t)B^T(t)P(t) + Q(t) \\ P(t_f) = F \end{cases}$$

• 此时:  $J^* = \frac{1}{2}x^T(t_0)P(t_0)x(t_0)$

$$\dot{x}^* = (A - BR^{-1}B^T P)x(t)$$

$$\mathcal{H} = \frac{1}{2}x^T Q x + \frac{1}{2}u^T R u + \lambda^T A x + \lambda^T B u$$

$$\frac{\partial \mathcal{H}}{\partial u} = R u + B^T \lambda = 0 \Rightarrow u^* = -R^{-1}B^T \lambda$$

$$\frac{\partial^2 \mathcal{H}}{\partial u^2} = R > 0, \text{ 故 } u^* \text{ 为最优控制.}$$

$$\begin{cases} \dot{x} = A x - B R^{-1} B^T \lambda \\ \dot{\lambda} = -Q x - A^T \lambda \end{cases}, \text{ 令 } \lambda = P x \Rightarrow \begin{cases} \dot{\lambda} = \dot{P} x + P \dot{x} \\ \Rightarrow \dot{x} = (A - B R^{-1} B^T P) x \end{cases}$$

↓  
最优闭环系统, 反馈阵  $R^{-1}B^T P$

$$\Rightarrow \begin{cases} \dot{\lambda} = (\dot{P} + P A - P B R^{-1} B^T P) x \\ \dot{\lambda} = -(Q + A^T P) x \end{cases}$$

$$\Rightarrow -\dot{P} - P A + A^T P - P B R^{-1} B^T P + Q = 0$$

• 黎卡提方程解的性质:  $u^*$  是线性反馈, 增益阵为  $K = R^{-1}B^T P$ .

iii)  $P$  是唯一且对称的矩阵;  $P$  是非负的;

iiii)  $P$  为时变阵, 实现十分困难.

例1.  $\dot{x} = \frac{1}{2}x + u, x(t_0) = x_0$

$$J = \int_{t_0}^{t_f} (\frac{1}{2}e^{-t}x^2 + 2e^{-t}u^2)dt$$

解  $A = \frac{1}{2}, B = 1, F = 0, Q = \frac{1}{2}e^{-t}, R = 2e^{-t}$

根据提方程  $-\dot{p}(t) = p(t) - \frac{1}{2}e^t p^2(t) + \frac{1}{2}e^{-t}, p(t_f) = 0$

为使方程线性化, 令  $\hat{x} = e^{-\frac{1}{2}t}x, \hat{u} = e^{-\frac{1}{2}t}u$ .

则 
$$\begin{cases} \dot{\hat{x}} = \hat{u} \\ -\dot{\hat{p}}(t) = -\frac{1}{2}\hat{p}^2(t) + \frac{1}{2}, \hat{p}(t_f) = 0 \end{cases}$$

$$\Rightarrow \hat{p}(t) = \frac{1 - e^{t-t_f}}{1 + e^{t-t_f}}$$

$$\Rightarrow \hat{u}^*(t) = -\frac{1}{2}e^{-\frac{1}{2}t} \hat{p}(t) x(t) \Rightarrow u^* = -\frac{1}{2}(1 - e^{t-t_f})(1 + e^{t-t_f})^{-1} x(t).$$

## 无限时间状态调节器

### 无限时间时变状态调节器

#### 无限时间时变状态调节器问题

$$\dot{x} = Ax + Bu$$

$$\min J = \int_{t_0}^{\infty} (x^T Q x + u^T R u) dt$$

若  $\{A, B\}$  完全能控, 则存在唯一最优控制

$$u^* = -R^{-1}B^T \bar{P} x$$

其中  $\bar{P} = \lim_{t \rightarrow \infty} P$

$$-\dot{P} = PA + A^T P - PBR^{-1}B^T P + Q$$

i)  $t_f \rightarrow \infty$  时, 存在系统能控性要求;

iii)  $F = 0$ , 因为  $t_f \rightarrow \infty$  的指标在实际中无意义.

### 无限时间定常状态调节器

#### 无限时间定常状态调节器:

$$\dot{x} = Ax + Bu$$

$$\min J = \frac{1}{2} \int_0^{\infty} (x^T Q x + u^T R u) dt$$

若  $\{A, B\}$  完全能控,  $\{A, D\}$  完全能观 ( $DD^T = Q$  且  $D$  任意), 则存在唯一最优控制.

$$u^* = -R^{-1}B^T \bar{P} x$$

$$\bar{P}A + A^T \bar{P} - \bar{P}BR^{-1}B^T \bar{P} + Q = 0, \bar{P} \text{ 为对称正定阵.}$$

该系统是稳定的,  $x^T \bar{P} x$  为 Lyapunov 函数.

$\{A, B\}$  可控  $\rightarrow$  最优解存在

$\{A, D\}$  可观  $\rightarrow$  系统稳定

• 有给定稳定度的状态调节器

$$\dot{x} = Ax + Bu$$

$$\min J = \frac{1}{2} \int_0^{\infty} e^{2\alpha t} (x^T Q x + u^T R u) dt$$

$u^*$  使  $J$  最小, 并使闭环系统极点小于  $-\alpha$ .

◦ 若  $[A, B]$  完全能控,  $[A, D]$  完全能观,  $DD^T = Q$ , 则存在唯一最优控制.

$$u^* = -R^{-1}B^T \bar{P} x$$

其中:  $\bar{P}(A + \alpha E) + (A^T + \alpha E)\bar{P} - \bar{P}BR^{-1}B^T\bar{P} + Q = 0$ .

此时系统渐近稳定的, 稳定度至少为  $\alpha$ .

## 离散状态调节器

• 离散状态调节器:

$$x(n+1) = A(n)x(n) + B(n)u(n)$$

$$\min J = \frac{1}{2} x^T(N)F x(N) + \frac{1}{2} \sum_{n=0}^{N-1} (x^T Q x + u^T R u)$$

$$F \geq 0, Q(n) \geq 0, R(n) > 0.$$

◦ 最优控制唯一,  $u^*(n) = -K(n)x(n)$

$$K(n) = (B^T(n)P(n+1)B(n) + R(n))^{-1} B^T(n)P(n+1)A(n)$$

$P(n)$  为离散黎卡提方程的对称非负定解:

$$\begin{cases} P(n) = (A(n) - B(n)K(n))^T P(n+1) (A(n) - B(n)K(n)) + K^T(n)R(n)K(n) + Q(n) \\ P(N) = F \end{cases}$$

◦ 离散黎卡提方程的迭代法.

$$\begin{cases} Z_1(n) = Q(n) + A^T(n)P(n+1)A(n) \\ Z_2(n) = B^T(n)P(n+1)A(n) \\ Z_3(n) = R(n) + B^T(n)P(n+1)B(n) \end{cases}$$

$$K(n) = Z_3^{-1}(n)Z_2(n), P(n) = Z_1(n) - Z_2^T(n)Z_3^{-1}(n)Z_2(n).$$

(i)  $P(N) = F$

(ii)  $n = N-1$ , 求  $Z_1(n), Z_2(n), Z_3(n)$

(iii) 计算  $K(n), P(n), u^*(n)$

(iv) 重复以上步骤.



### 3. 输出调节器

#### 有限时间输出调节器

· 有限时间输出调节器:

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx \end{cases}$$

$$\min J = \frac{1}{2} y^T(t_f) F y(t_f) + \frac{1}{2} \int_{t_0}^{t_f} (y^T Q y + u^T R u) dt$$

$F$  为对称非负定常阵,  $Q$  为非负定阵,  $R$  为正定阵.

◎ 若  $\{A, C\}$  完全可观, 则存在唯一最优控制:

$$u^*(t) = -R^{-1}(t) B^T(t) P(t) x(t)$$

$P(t)$  满足黎卡提方程

$$\begin{cases} -\dot{P}(t) = P(t)A(t) + A^T(t)P(t) - P(t)B(t)R^{-1}(t)B^T(t)P(t) + C^T(t)Q(t)C(t) \\ P(t_f) = C^T(t_f) F C(t_f) \end{cases}$$

#### 无限时间输出调节器

· 无限时间定常输出调节器:

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx \end{cases}$$

$$\min J = \frac{1}{2} \int_0^\infty (y^T Q y + u^T R u) dt$$

◎ 若  $\{A, B\}$  完全能控,  $\{A, C\}$  完全能观, 对于  $DD^T = C^T Q C$  的  $D$ ,  $\{A, D\}$  完全能观,

则

$$u^*(t) = -R^{-1} B^T \bar{P} x$$

$\bar{P}$  满足黎卡提方程:  $\bar{P}A + A^T \bar{P} - \bar{P}B R^{-1} B^T \bar{P} + C^T Q C = 0$

$\bar{P}$  为正定对称常阵.

#### 输出反馈次优调节器

· 输出反馈次优调节器

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx \end{cases}$$

$$\min J = \frac{1}{2} \int_0^\infty (y^T Q y + u^T R u) dt$$

◎ 取正定常阵  $P$ ,  $\bar{Q}$  ( $\bar{Q} = Q + C^T K^T R K C$ )

次优控制

$$u^o(t) = -K C x(t)$$

$P, K$  满足 Lyapunov 方程:

$$(A - BKC)^T P + P(A - BKC) = -(\bar{Q} + C^T K^T R K C)$$

◎ 次优指标

$$J^o = \frac{1}{2} x^T(0) P x(0)$$

$$\frac{\partial J^*(k)}{\partial k} = 0$$

## 离散输出调节器

· 离散输出调节器: 
$$\begin{cases} x(n+1) = A(n)x(n) + B(n)u(n) \\ y(n) = C(n)x(n) \end{cases}$$

$$\min J = \frac{1}{2} y^T(N) F y(N) + \frac{1}{2} \sum_{n=0}^{N-1} (y^T Q y + u^T R u)$$

○ 若  $\{A, C\}$  完全可观, 则有唯一最优控制:

$$u^*(n) = -k(n)x(n)$$

$P(n)$  满足黎卡提方程:

$$\begin{cases} P(n) = (A(n) - B(n)k(n))^T P(n+1) (A(n) - B(n)k(n)) + k^T(n) R(n) k(n) + C^T(n) Q(n) C(n) \\ P(N) = C^T(N) F C(N) \end{cases}$$

○ 设计步骤: (i) 检验系统可观性;

(ii)  $k = N$ , 解  $P(N) = C^T(N) F C(N)$

(iii)  $k = N-1$ , 解

$$\begin{cases} Z_1(n) = C^T(n) Q(n) C(n) + A^T(n) P(n+1) A(n) \\ Z_2(n) = B^T(n) P(n+1) A(n) \\ Z_3(n) = R(n) + B^T(n) P(n+1) B(n) \\ k(n) = Z_3^{-1}(n) Z_2(n) \\ P(n) = Z_1(n) - Z_2^T(n) Z_3^{-1}(n) Z_2(n) \end{cases}$$

(iv) 重复以上步骤, 向前迭代.

#### 4. 跟踪系统

##### 有限时间时变跟踪系统

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx \end{cases}$$

$e(t) = y_d(t) - y(t)$ ,  $y_d(t)$  为期望输出.

$$\min J = \frac{1}{2} e^T(t_f) F e(t_f) + \frac{1}{2} \int_{t_0}^{t_f} (e^T(t) Q(t) e(t) + u^T(t) R(t) u(t)) dt$$

若  $\{A, C\}$  可观, 存在唯一最优控制:

$$u^* = -R^{-1} B^T (Px - g)$$

$P$  为对称非负矩阵, 满足黎卡提方程

$$\begin{cases} -\dot{P} = PA + A^T P - PBR^{-1}B^T P + C^T Q C \\ P(t_f) = C^T(t_f) F C(t_f) \end{cases}$$

$g$  为伴随变量:

$$\begin{cases} -\dot{g} = (A - BR^{-1}B^T P)g + C^T Q C \\ g(t_f) = C^T(t_f) F y_d(t_f) \end{cases}$$

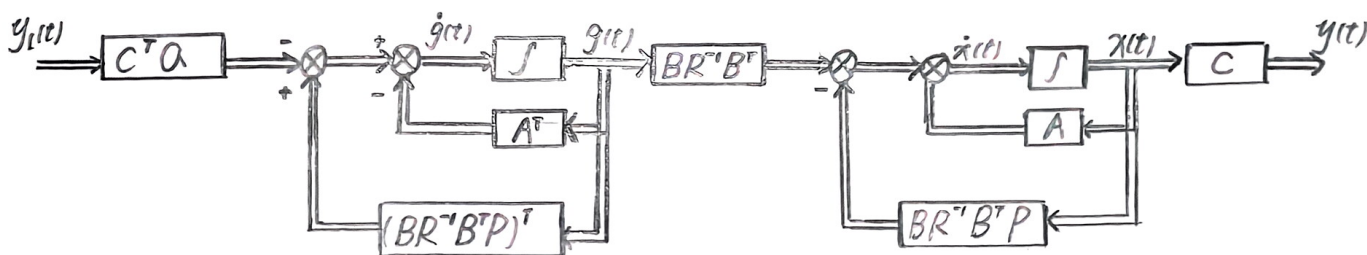
① (i) 最优跟踪系统的反馈结构与期望输出无关,  $P^T B^T g$  为前馈项.

(ii) 最优跟踪系统的动态性能与期望输出无关.

(iii) 最优跟踪系统现在值与期望输出将来值有关.

将  $y_d(t)$  视为某典型变化律  $\rightarrow$  性能取决于  $y_d(t)$  与  $y(t)$  符合程度  
将  $y_d(t)$  视为随机信号  $\rightarrow$  性能在平均意义下最优.

(iv) 离线求出  $P$  的值, 再求解  $g$ .



##### 无限时间定常跟踪系统

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx \end{cases}$$

$e(t) = \hat{y}_d - y(t)$ ,  $\hat{y}_d$  为期望常值输出

$$\min J = \frac{1}{2} \int_0^\infty (e^T Q e + u^T R u) dt$$

若  $\{A, B\}$  完全能控,  $\{A, C\}$  完全能观, 则近似最优控制

$$\hat{u} = -R^{-1} B^T \hat{P} x + R^{-1} B^T \hat{g}$$

$\hat{P}$  满足代数黎卡提方程

$$\hat{P}A + A'\hat{P} - \hat{P}BR^{-1}B'\hat{P} + C'QC = 0$$

$\hat{g}$  为常向量:

$$\hat{g} = (\hat{P}BR^{-1}B' - A')^{-1} C'Q \hat{y}_i$$